

FlockML

Distributed compute infrastructure for sovereign artificial intelligence workloads.

Seed Investment Presentation

Supratim Dhara · Founder & Chief Systems Architect

supratimdev.qzz.io/flock-ml

AI Compute Is Becoming a Bottleneck

The modern AI boom is encountering severe structural constraints across cost, access, and infrastructure centralization.

01. CENTRALIZATION

AI infrastructure is heavily concentrated in three hyperscale clouds. Developers face rigid pricing, forced minimum instance sizes, and arbitrary compute quotas.

02. EXPONENTIAL COST

Running multi-billion parameter models in centralized data centers requires enormous CapEx for specialized cooling, power grids, and hardware depreciation.

03. SOVEREIGN RISK

Nations and enterprises are sending sensitive proprietary datasets to offshore data centers, creating massive data sovereignty, security, and privacy vulnerabilities.

THE CORE PROBLEM

Centralized data centers cannot sustainably scale to meet the exponential global demand for AI inference and training.

Why Now? Three Structural Shifts

Three technological convergence points make decentralized edge compute viable for the first time in history.

01. WEBGPU & WASM SIMD

Modern browsers and edge runtimes now expose bare-metal GPU acceleration (WebGPU) and 128-bit vector math (Wasm SIMD) with zero native driver installation.

02. 1.58-BIT QUANTIZATION

Ternary quantization (BitNet b1.58) replaces power-hungry floating-point multiplications with integer additions, reducing model memory footprints by 80.2%.

03. SILICON SATURATION

Billions of consumer laptops, Apple Silicon chips, and telecom set-top boxes sit idle over 18 hours a day, wired to high-speed gigabit fiber and wall power.

MARKET TIMING

The combination of WebGPU, BitNet quantization, and pervasive fiber makes a zero-CapEx decentralized compute grid technically and economically inevitable.

A Cloud Experience Over a Distributed Compute Grid

FlockML abstracts millions of heterogeneous devices into a single, programmable cloud API.

WHAT DEVELOPERS SEE

DEVELOPER API

- **Drop-In API:** Submit standard OpenAI or PyTorch requests and receive outputs instantly.
- **Familiar Primitives:** Deployments, inference endpoints, logs, and telemetry with standard SDKs.
- **Zero Cluster Setup:** No manual server management, SSH keys, or CUDA driver maintenance.

WHAT FLOCKML ORCHESTRATES

RUNTIME MESH

- **Dynamic Discovery:** Automatically matches workloads to active nodes across the network.
- **Smart Tensor Sharding:** Partitions large models across heterogeneous device RAM limits.
- **Sub-5ms Failover:** Dynamic work-stealing automatically reroutes dropped nodes with zero data loss.

THE DEVELOPER PROMISE

The developer thinks about the workload. FlockML handles the infrastructure underneath.

How the System Works: The Execution Layer

Four specialized subsystems turn untrusted edge hardware into a fault-tolerant compute cluster.

01. FLOCKML WORKER

SANDBOX

Zero-install runtime in WebAssembly SIMD and WebGPU. Executes tensor matrix math on consumer chips with zero host privilege.

02. DISTRIBUTED COORDINATOR

TOPOLOGY

High-throughput orchestration layer tracking node telemetry, latency, and RAM to shard model weights dynamically across the network.

03. zk-VERIFICATION

SUB-0.1MS

Zero-knowledge proof-of-compute commitments evaluate instantly without re-running weights, rejecting Byzantine or corrupt updates.

04. FAULT-TOLERANT FAILOVER

SUB-5MS

Work-stealing P2P ring scheduler. If a node drops, neighboring machines seamlessly pick up the pipeline stage with zero token loss.

SYSTEM ARCHITECTURE

Untrusted edge devices are transformed into an enterprise-grade compute substrate through deterministic verification and sub-5ms failover.

Technical Proof: BitNet 1.58-Bit Quantization

Mathematical breakthroughs in ternary weight representation make edge AI faster, cheaper, and radically efficient.

80.2% VRAM REDUCTION

Ternary weights $\{-1, 0, +1\}$ pack 16 parameters into a single 32-bit register. A 70B parameter model that normally requires 140GB VRAM now fits comfortably across everyday edge memory pools.

INTEGER ADDITIONS

Replaces energy-heavy floating-point matrix multiplications (FP16/FP32) with lightweight integer additions, slashing power consumption by 85% on standard consumer CPUs and integrated GPUs.

SUB-0.1MS VERIFICATION

Coordinator validates mathematical integrity through coordinate-wise hash commitments without re-executing inference, maintaining line-rate execution across thousands of nodes.

THE QUANTIZATION LEAP

BitNet 1.58-bit turns consumer chips into high-speed inference engines, eliminating the requirement for expensive \$30,000 datacenter GPUs.

The Economics: Hyperscale vs. Decentralized Compute

By eliminating data center CapEx, we fundamentally restructure the cost of AI compute while retaining software margins.

HYPERSCALE CLOUD

~\$32.00 (~₹2,650)

MINIMUM HOURLY BILL (FORCED 8x A100 NODE)

Why it is expensive:

- Forced to rent massive 8-GPU nodes with rigid minimums
- Datacenter real estate, HVAC cooling, and power overhead
- Corporate margin stacking across infrastructure layers
- Continuous rapid hardware depreciation

FLOCKML DECENTRALIZED

~\$10.00 (~₹830)

PER HOUR (FLEXIBLE DISTRIBUTED CLUSTER)

Why it is disruptive:

- Developers save ~70% compared to legacy cloud minimums
- Zero hardware CapEx or datacenter cooling costs for FlockML
- Compute providers earn high yield on sunk-cost hardware
- **78%+ Gross Margins (Pure software take-rate)**

When compute costs drop by 70%+, developers can deploy entirely new classes of AI applications that were previously economically unviable.

Developer Value & Incremental Migration

We eliminate enterprise switching friction through a 4-step Trojan-horse adoption funnel.

WHY DEVELOPERS MIGRATE

VALUE PILLARS

01. 70% COST REDUCTION

Bypass forced 8-GPU cloud minimums and pay strictly for active matrix cycles executed.

02. INSTANT CAPACITY

Access millions of available edge and enterprise cores with zero GPU waitlists or quotas.

03. ZERO LOCK-IN

Standard REST/OpenAI interfaces make routing workloads to FlockML take under 10 lines of code.

THE 4-STEP ADOPTION FUNNEL

LOW FRICTION

- 01 Add Secondary Target:** Keep AWS stack intact; plug in FlockML API key in 5 minutes.
- 02 Route Batch & Eval:** Offload non-critical offline fine-tuning, embeddings, and synthetic data tests.
- 03 Dynamic Routing:** SDK automatically routes traffic to wherever compute is cheapest.
- 04 Scale Production:** As confidence builds, production inference organically shifts onto FlockML.

MIGRATION PRINCIPLE

Do not ask developers to rewrite their world on Day 1. Give them a zero-risk, high-margin execution target.

The Execution Layer for Distributed Compute

FlockML abstracts millions of heterogeneous devices into a single, programmable cloud for AI workloads.

TRADITIONAL VS. FLOCKML

PARADIGM SHIFT

- **Hyperscale Cloud:** Own physical server racks → Rent expensive fixed instances.
- **GPU Marketplaces:** Aggregate individual servers → Manual SSH and fragmented hardware.
- **FlockML:** Pure software orchestration layer → Developers interact with a standard API while our runtime handles sharding, failover, and execution underneath.

WHAT THE RUNTIME HANDLES

ZERO OVERHEAD

- **Dynamic Tensor Sharding:** Automatically partitions 70B & 405B models across varied device RAM limits.
- **Cryptographic Verification:** Sub-0.1ms zk-SNARK gates validate matrix computation.
- **Self-Healing Failover:** Sub-5ms rerouting ensures zero pipeline stall if a device disconnects.

CORE MOAT

FlockML does not need to own the world's compute. We become the software layer through which the world's available compute is executed.

Where FlockML Sits

Positioned at the intersection of enterprise cloud usability and zero-CapEx decentralized compute.

MODEL	INFRASTRUCTURE	PRICING / CAPEX	DEVELOPER UX	FAULT TOLERANCE
Hyperscale Cloud	Centralized Datacenters	High (\$32/hr minimums)	Standard Cloud APIs	Centralized Failover
GPU Marketplaces	Fragmented Servers	Medium (Spot rates)	Complex (Manual SSH)	Manual Reconnect
Crypto DePINs	Consumer Rigs (Tokenized)	Volatile Token Gas	Cryptic Web3 Wallets	High Failure Rates
FlockML	Decentralized Edge Mesh	Disruptive (\$10/hr Flat)	Standard REST / SDK	Sub-5ms Self-Healing

STRATEGIC MOAT

We do not ask enterprises to manage crypto wallets or raw SSH keys. We offer a Tier-1 cloud developer experience on top of a zero-CapEx distributed compute grid.

The Compute Supply Engine: Office Laptops to 100M Telecom Boxes

How FlockML aggregates petabytes of compute capacity with zero infrastructure capital expenditure.

01. ENTERPRISE FLEETS

B2B SUPPLY

Thousands of corporate and government laptops (MeitY, CESC, banks) sit idle from 7 PM to 7 AM. Organizations run internal AI models across their own hardware grid with zero data leaving the perimeter.

02. TELECOM SET-TOP BOXES

100M+ SCALE

30M+ Reliance Jio & Airtel 4K boxes connected to fiber and wall power, idle 20 hours/day. Single over-the-air Wasm injection creates a **150 PFLOPs national living-room supercomputer**.

03. UTILIZATION-BASED YIELD

ZERO IDLE BURN

FlockML pays providers strictly for active matrix cycles executed. Telcos earn new high-margin enterprise revenue, while consumers receive broadband bill credits.

SCALE MULTIPLIER

10M Jio/Airtel Set-Top Boxes + Wasm SIMD = 10 Petabytes VRAM & 150 PFLOPs Compute with \$0 Cloud CapEx.

A Compute Cloud Business Model

High software gross margins through usage-based compute billing and enterprise private deployments.

01. USAGE COMPUTE API

Developers pay per token or compute-second (\$10/hr cluster equivalent).
FlockML captures a 75% to 80% software margin after node provider payouts.

02. ENTERPRISE PRIVATE GRID

Annual software licensing (\$100k-\$500k ARR) for enterprises and banks to turn internal office machines into a private, air-gapped sovereign AI training cluster.

03. SOVEREIGN AI DEPLOYMENTS

Government and national AI grid contracts (IndiaAI, MeitY, Gulf Sovereigns) deploying national model training infrastructure with full data localization.

HIGH-MARGIN SOFTWARE ECONOMICS

FlockML operates with pure software gross margins (78%+) because we never purchase, house, or depreciate physical server hardware.

The Next 24 Months: Proof → Product → Network → Scale

A disciplined, milestone-driven execution plan focused on technical validation and enterprise distribution.

PHASE 1: PROTOCOL & BENCHMARKS

MONTHS 1-6

- Deploy WebGPU & Wasm SIMD kernels for Llama-3 70B.
- Complete 500-node MeitY & enterprise private grid pilot.
- Release developer SDK & CLI with drop-in OpenAI parity.

PHASE 2: TELECOM PILOT & PUBLIC API

MONTHS 7-12

- Partner with Indian telecom for 50,000 set-top box pilot.
- Launch public developer compute API with usage billing.
- Scale network to 10,000 active nodes with sub-5ms failover.

PHASE 3: ENTERPRISE COMMERCIALIZATION

MONTHS 13-18

- Sign 20+ enterprise customers across banking and defense.
- Expand telecom footprint to 1M+ connected living rooms.
- Reach \$3M ARR with positive software unit economics.

PHASE 4: GLOBAL SOVEREIGN GRIDS

MONTHS 19-24

- Deploy sovereign AI grid infrastructure in UAE & Gulf markets.
- Scale network beyond 10M active edge devices globally.
- Establish FlockML as the global distributed execution layer.

EXECUTION MILESTONES

From technical proof to millions of distributed nodes delivering sovereign AI compute.

The Ask: Seed Round Capital Allocation

Raising \$20.0M (~₹168 Crore) Seed to scale our distributed systems engineering team and accelerate telecom partnerships.

ROUND PARAMETERS

SEED ROUND

\$20.0M (~₹168 Cr)

TARGET SEED RAISE

\$80M Pre / \$100M Post (~₹670 Cr / ~₹840 Cr)

VALUATION FRAMEWORK

24 - 30 Months Runway

PATH TO SERIES A & SOVEREIGN SCALE

Co-investors: Deep Tech, Infrastructure & Sovereign Funds

CAPITAL ALLOCATION

DEPLOYMENT

40% Core Distributed Systems **\$8.0M** (~₹67.2 Cr)

Wasm SIMD kernels, WebGPU compute pipelines, zk-SNARK verifiers.

20% Telecom & Enterprise BD **\$4.0M** (~₹33.6 Cr)

Telecom set-top box rollouts, MeitY pilots, enterprise private grids.

15% Developer Ecosystem **\$3.0M** (~₹25.2 Cr)

Global developer SDK, documentation, hackathons, and grants.

10% Security & Compliance **\$2.0M** (~₹16.8 Cr)

SOC 2 Type II, DPDP Act 2023 compliance, cryptographic audits.

15% Treasury & Contingency Buffer **\$3.0M** (~₹25.2 Cr)

Emergency runway extension & operational liquidity safety cushion.

INVESTMENT CATALYST: Capital will establish FlockML as the category leader in zero-CapEx distributed AI infrastructure.

Why FlockML Wins: Investment Summary

A category-defining software infrastructure layer capturing the global transition to distributed, sovereign AI compute.

01. ZERO-CAPEX SCALE

78% MARGIN

Zero server purchases, cooling, or datacenter real estate. Unlocks petabytes of compute from idle enterprise laptops and 100M telecom set-top boxes.

02. MATHEMATICAL MOAT

PATENT-PENDING

BitNet 1.58-bit ternary quantization (80.2% VRAM reduction) paired with sub-0.1ms zk-SNARK verification and sub-5ms self-healing failover.

03. LOW-FRICTION ADOPTION

DROP-IN API

Standard OpenAI/PyTorch drop-in API. Developers offload batch fine-tuning and evaluation with zero code rewrites, saving ~70% on compute bills.

04. SOVEREIGN DEFENSE

INDIAAI / MEITY

100% on-soil air-gapped data localization for government, banking, and defense. Eliminates dependence on foreign hyperscaler monopolies.

SUPRATIM DHARA • FOUNDER & CHIEF SYSTEMS ARCHITECT

supratim@flockml.com • supratimdev.qzz.io/flock-ml • Kolkata, India

READY FOR QUESTIONS →